

FoodHub: Data Analysis and Solutions

FoodHub Project for AI & ML by Erick Quintero

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Contents & Agenda

- Executive Summary
- Business Problem Overview and Solution Approach
- Data Overview
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- EDA - Multivariate Analysis

Executive Summary - Conclusions

- While looking at the data American cuisine is the most popular cuisine that is ordered. 30.71% of orders on the weekend and 30.89% on weekdays
- Korean and Vietnamese cuisine are the fastest to prepare
- Vietnamese cuisine has the lowest mean order cost around \$12
- 736 orders are not rated therefore limiting the available ratings that can be examined
- 10.54% of the orders take over 60 minutes to be delivered to the customer thus negatively affecting the ratings

Executive Summary recommendations

- Knowing that Korean and Vietnamese food have the lowest median preparation time, the company should have promotions to encourage people to purchase restaurants with this cuisine to keep lower deliver times during peak hours
- Introduce loyalty rewards program and add special offers to one time customers to increase the number of repeat customers. You can also target customers who have been inactive for over 45 days and send them offers to order food
- Offer to waive delivery fees for the next order of a customer if they give a rating on their order. This will give us more data to work with to accurately check which restaurants are performing better than others

Data Overview: Questions 1-3

- This data consists of 1898 rows and columns
- Columns: 'order_id', 'customer_id', 'food_preparation_time' and 'delivery_time' are all integers
- Columns: 'restaurant_name', 'cuisine_type', 'day_of_the_week', 'rating' are all objects
- Column 'cost_of_the_order' is a float
- There are no missing values in the data

Data Overview: Questions 4&5

Statistical summary of the data shows us the minimum, average and maximum time it takes for food to be prepared once an order is placed.

- Minimum time = 20 minutes
- Average time = ~27 minutes
- Maximum time = 35 minutes

There are also 736 orders not rated

Univariate Analysis

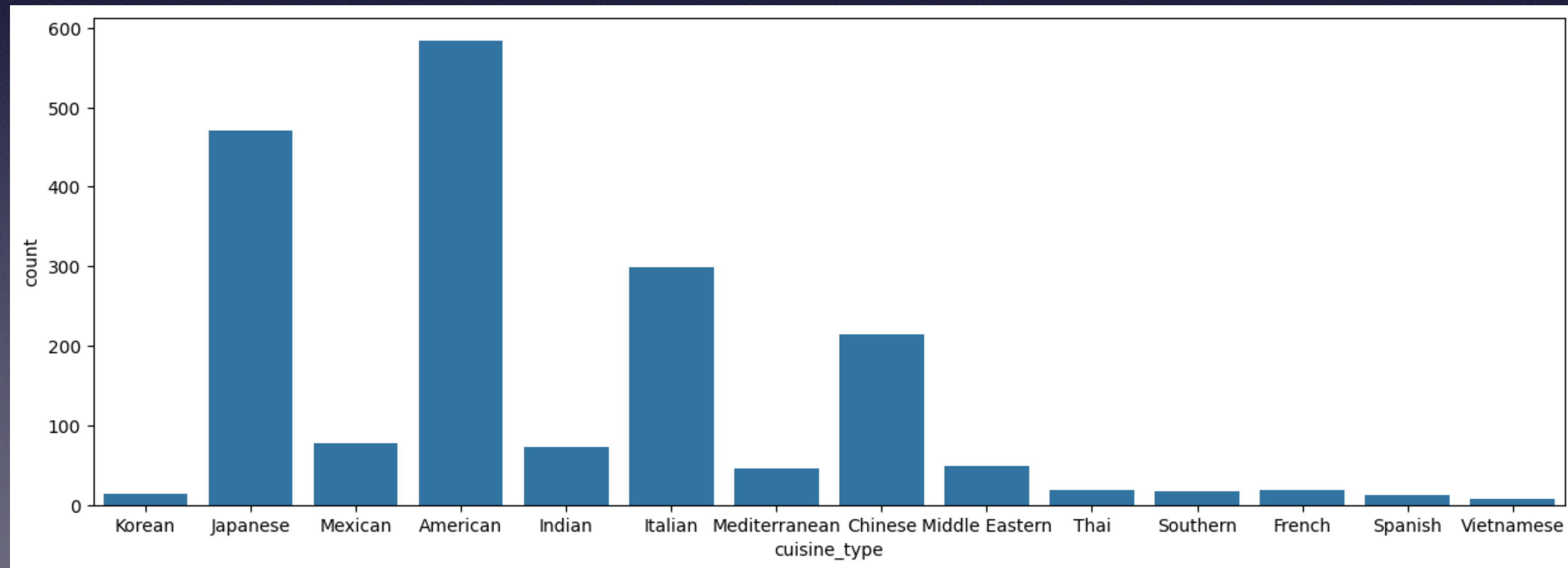
We will look into each variable and dive deeper into each one of them. Below is a list of the number of unique datapoints in each variable:

- 1898 unique Order ID's
- 1200 unique Customer ID's
- 178 unique Restaurant names
- 14 unique Cuisine types
- 2 unique Day of the weeks
- 4 unique rating types

We will dive deeper and visualize some of these variables.

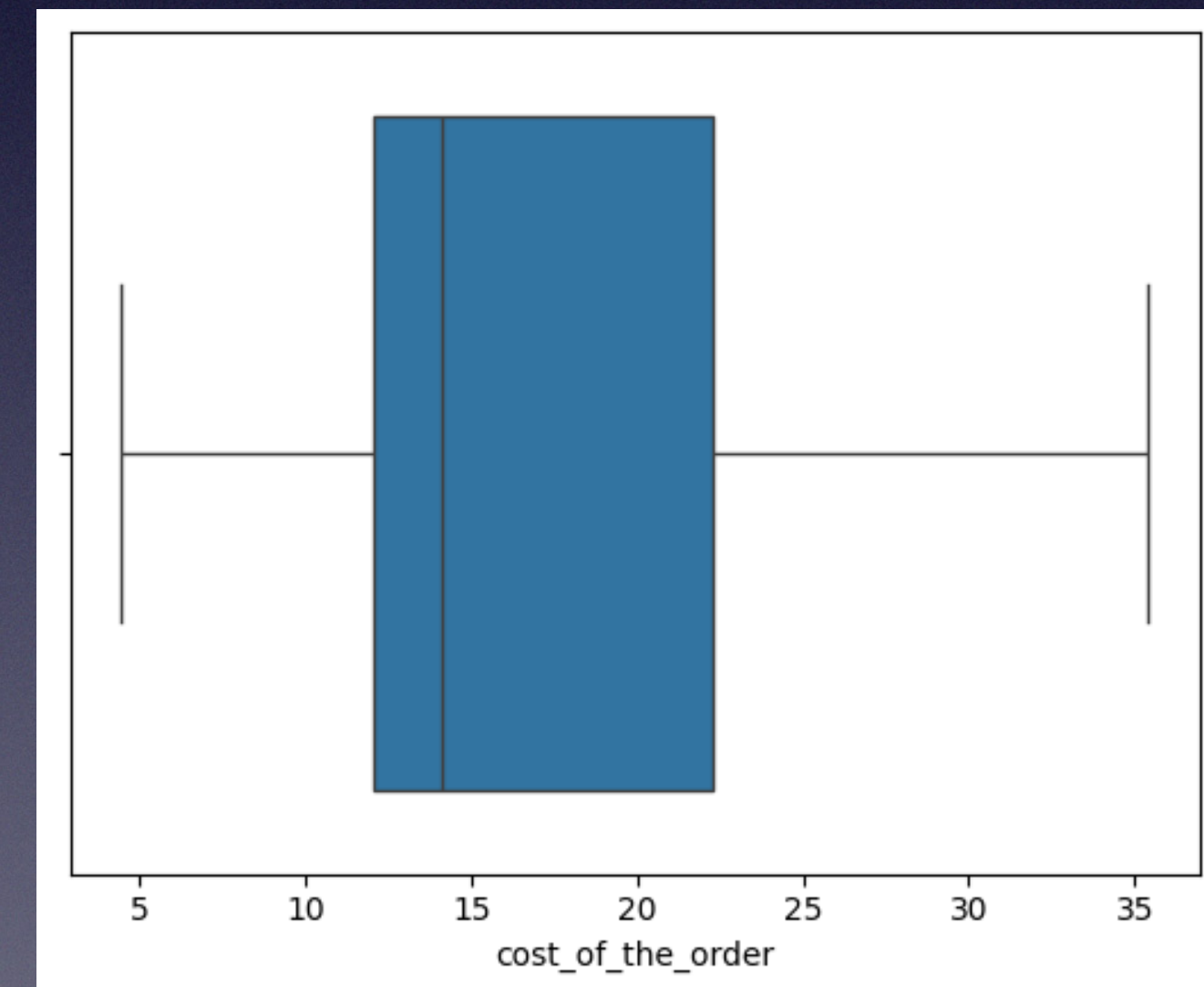
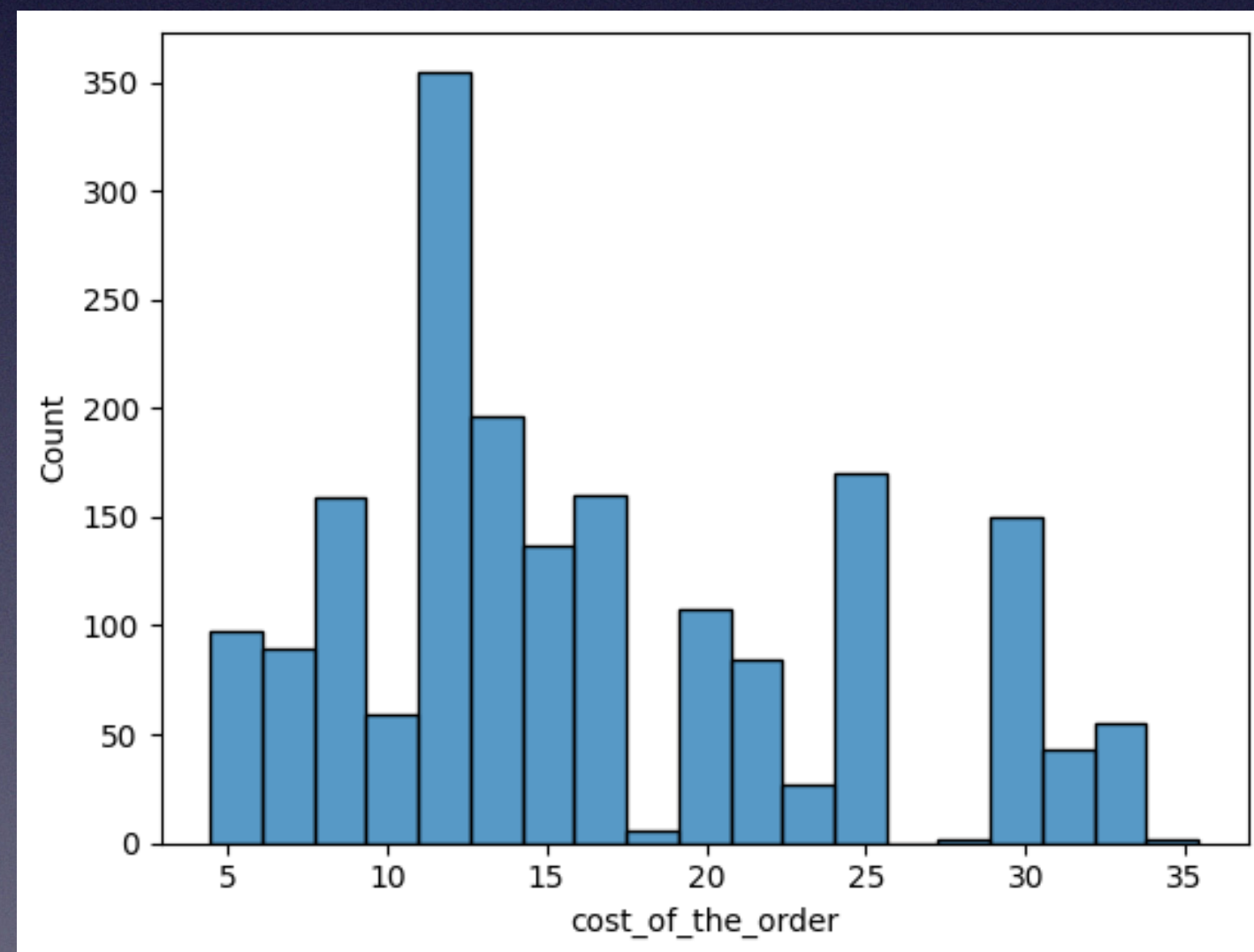
Cuisine Type

Below we see the 14 unique cuisine types and how frequently they get ordered.



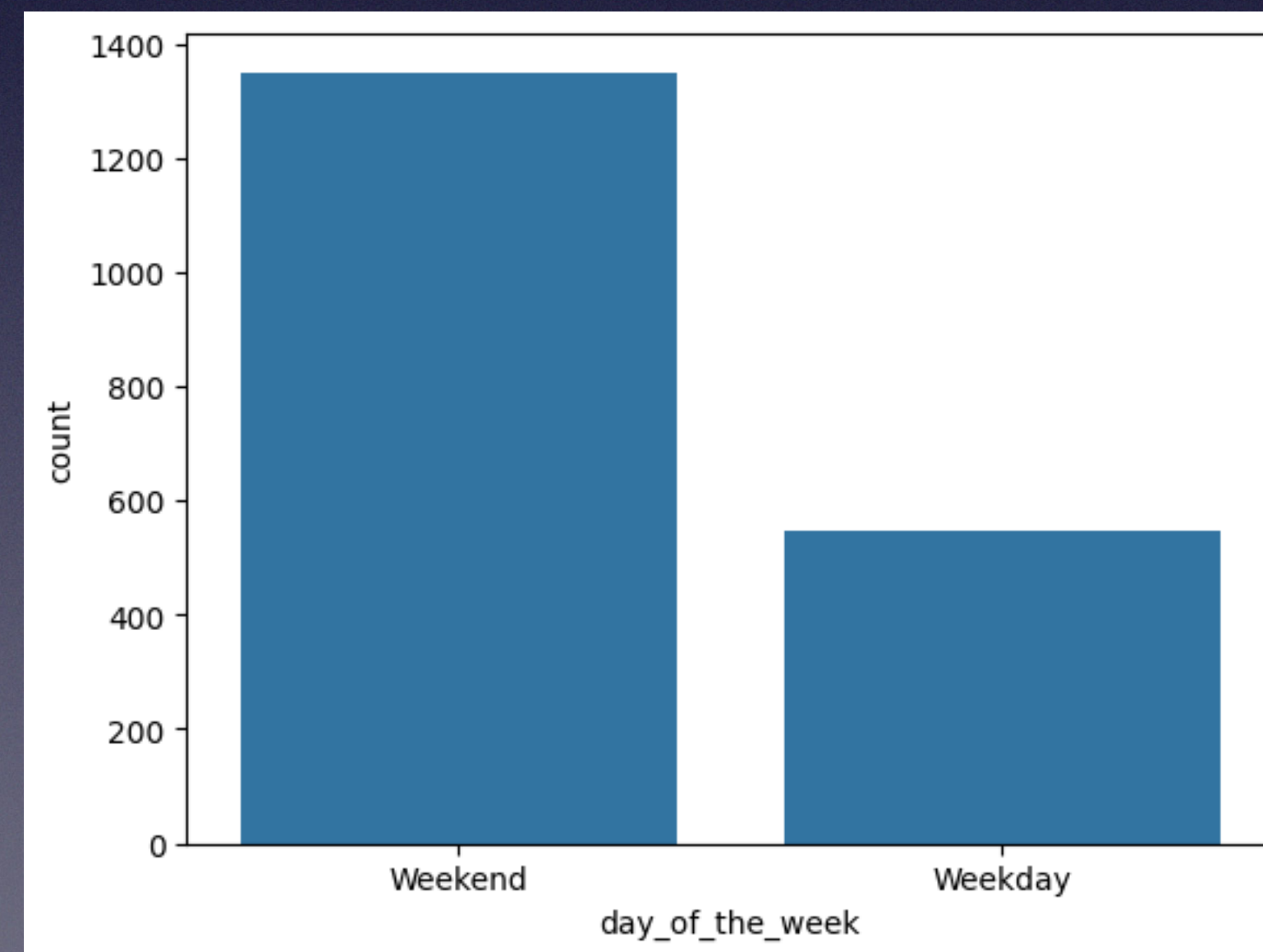
Cost of The Order

Below you can see the Histogram and Box plot. The Histogram is great to use to see the distribution of the cost of orders and see how often different priced orders are placed. The Box plot condenses the data in a way where you can analyze it in a more structured manner. By that I mean you are able to see the mean cost of orders are right under \$15 and the majority of the cost of orders are between ~\$12 & ~\$23. It also shows the the minimum cost of an order placed is about ~\$4 and the maximum cost of an order placed is ~\$36.



Day of The Week

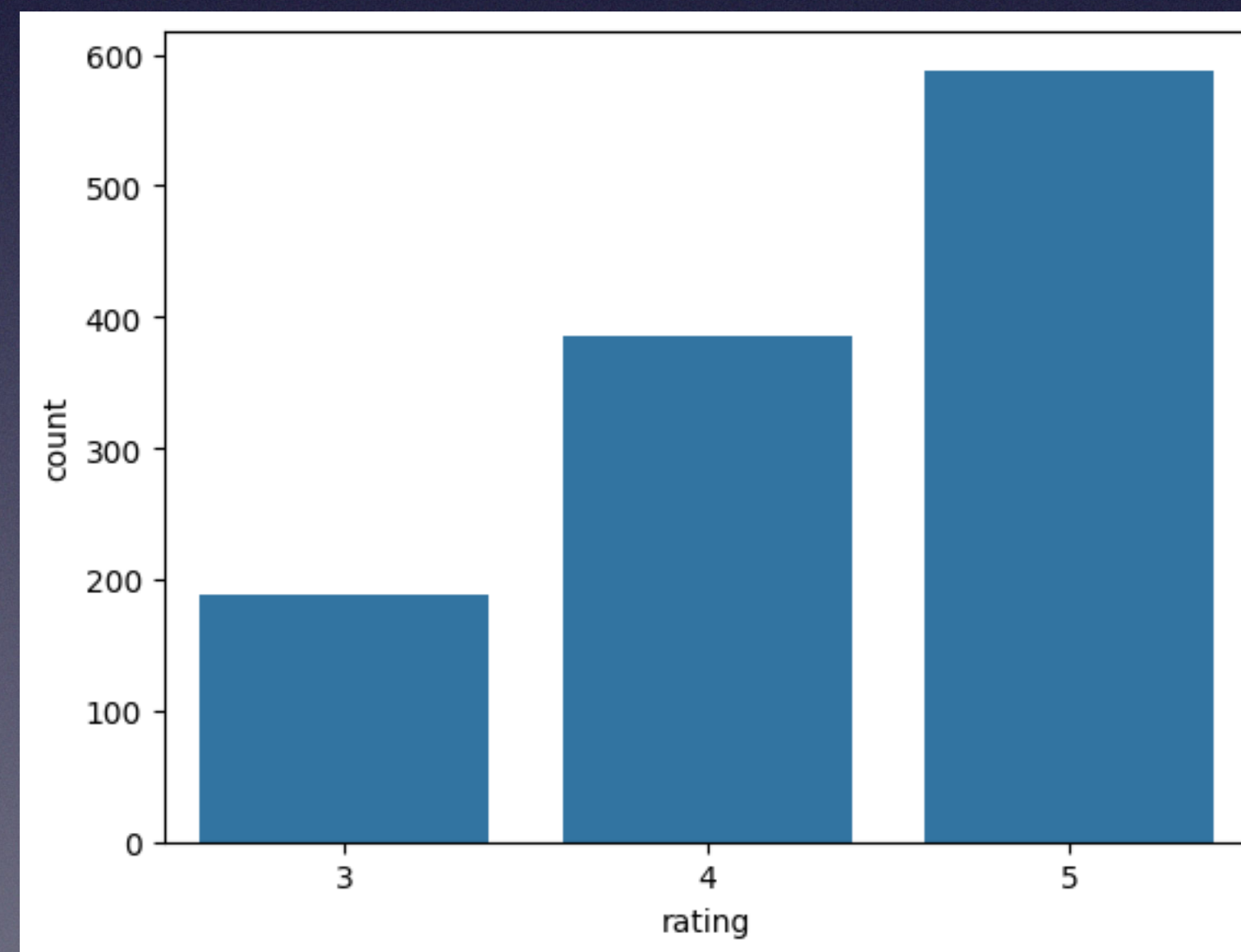
We see there are ~1350 orders placed on the weekend and ~550 placed on the weekday.



Ratings

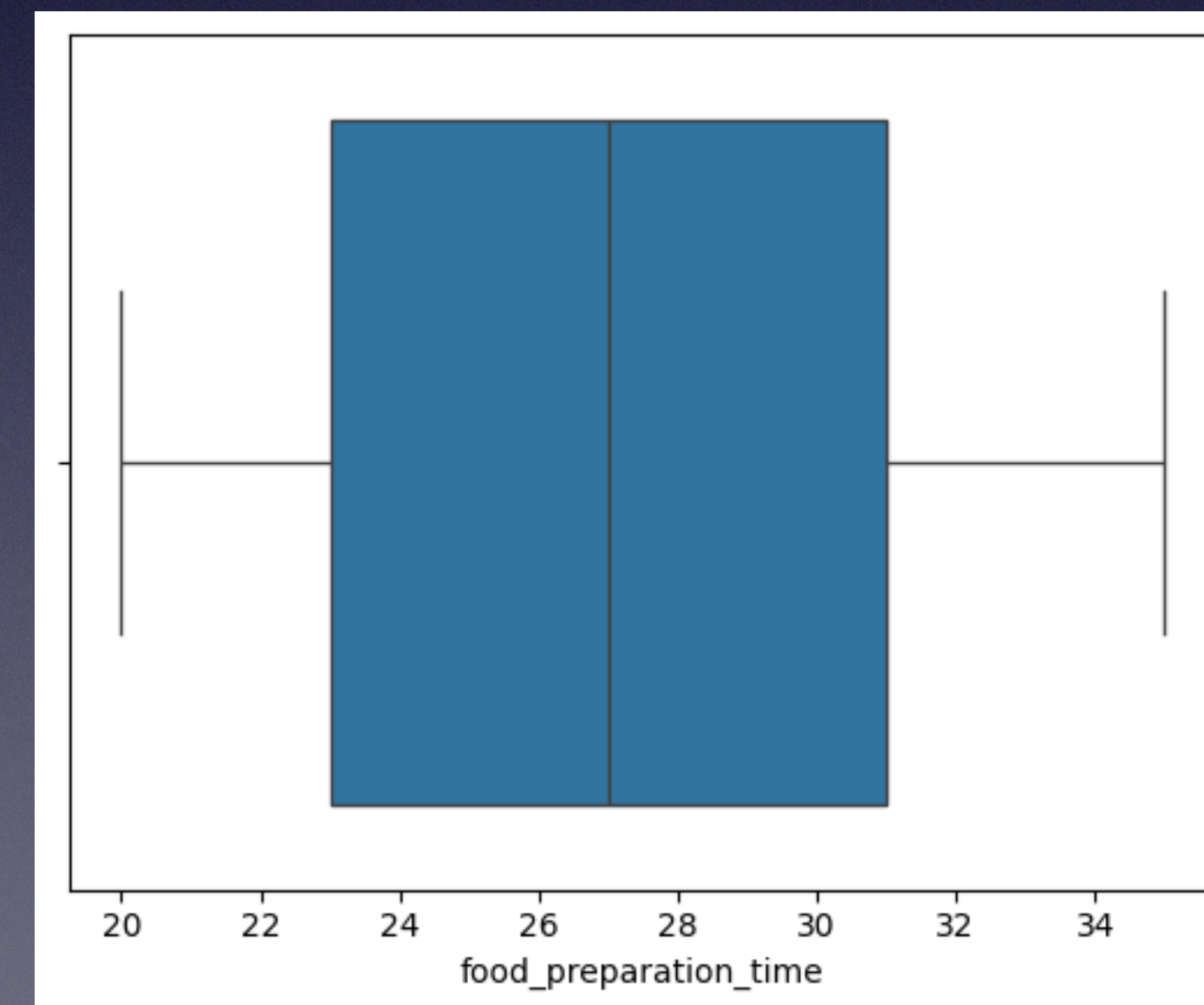
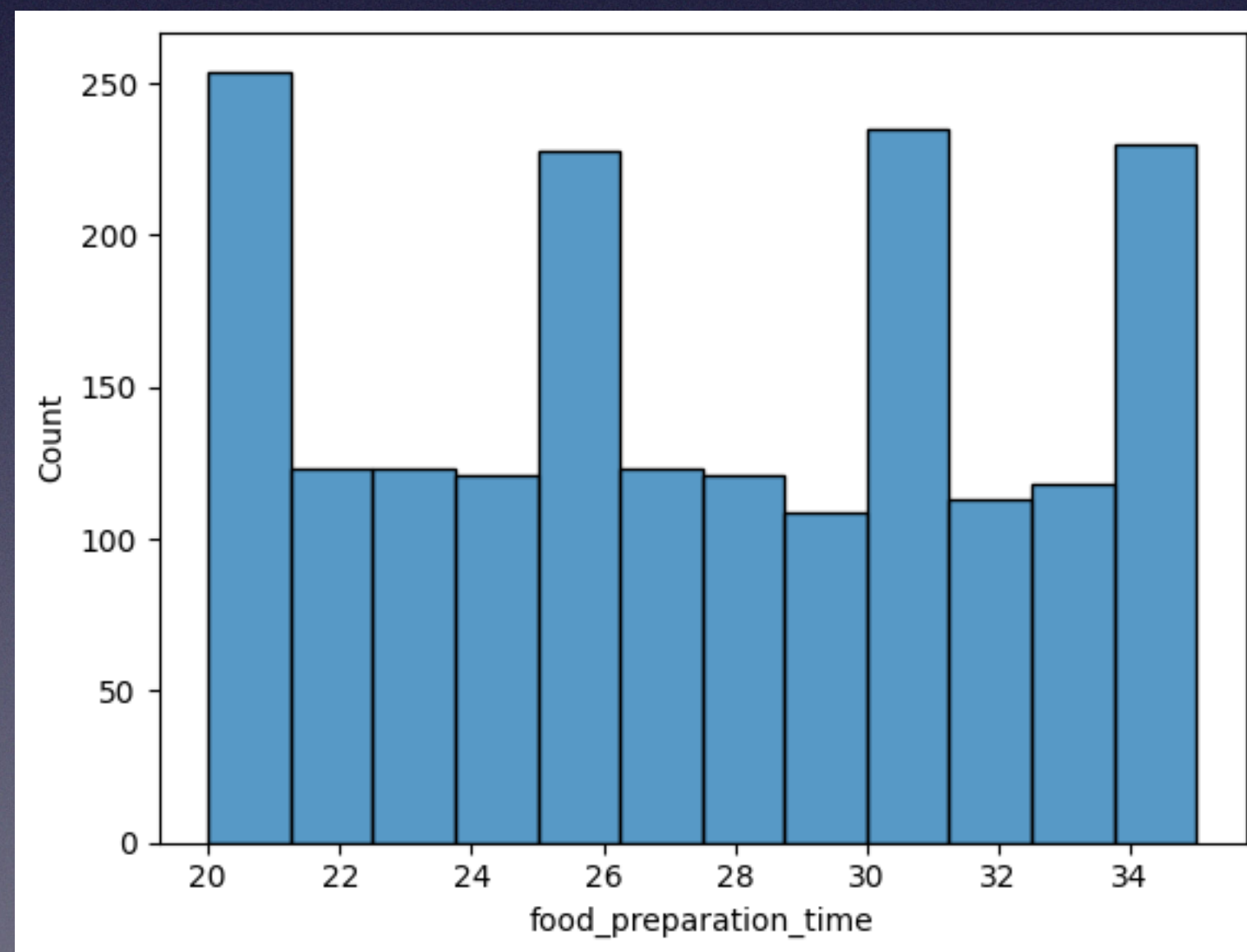
In the graph below we can see the unique values of each rating given:

Rating 3 has ~200 ratings, rating 4 has ~400 ratings and rating 5 has ~600 rating.



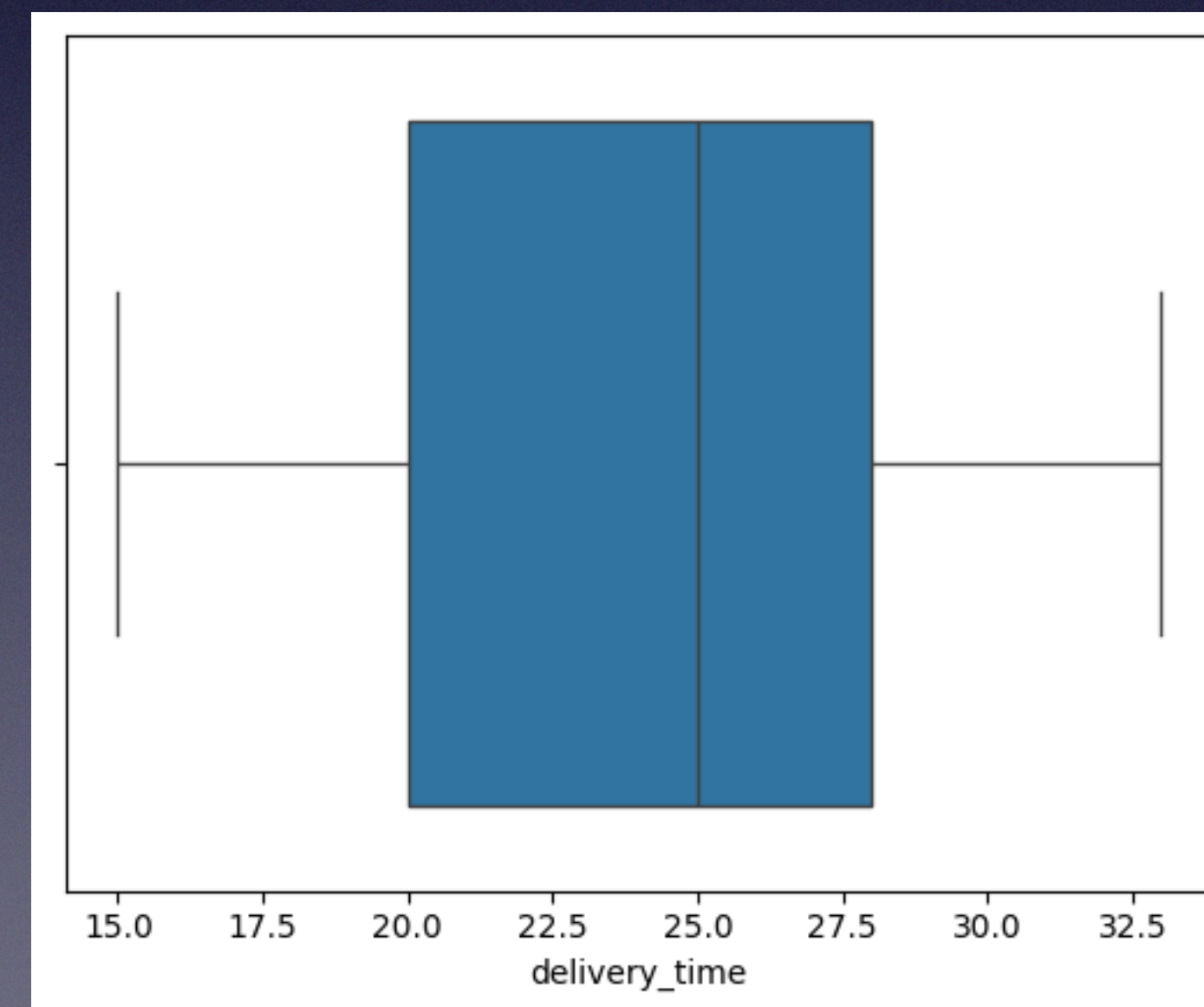
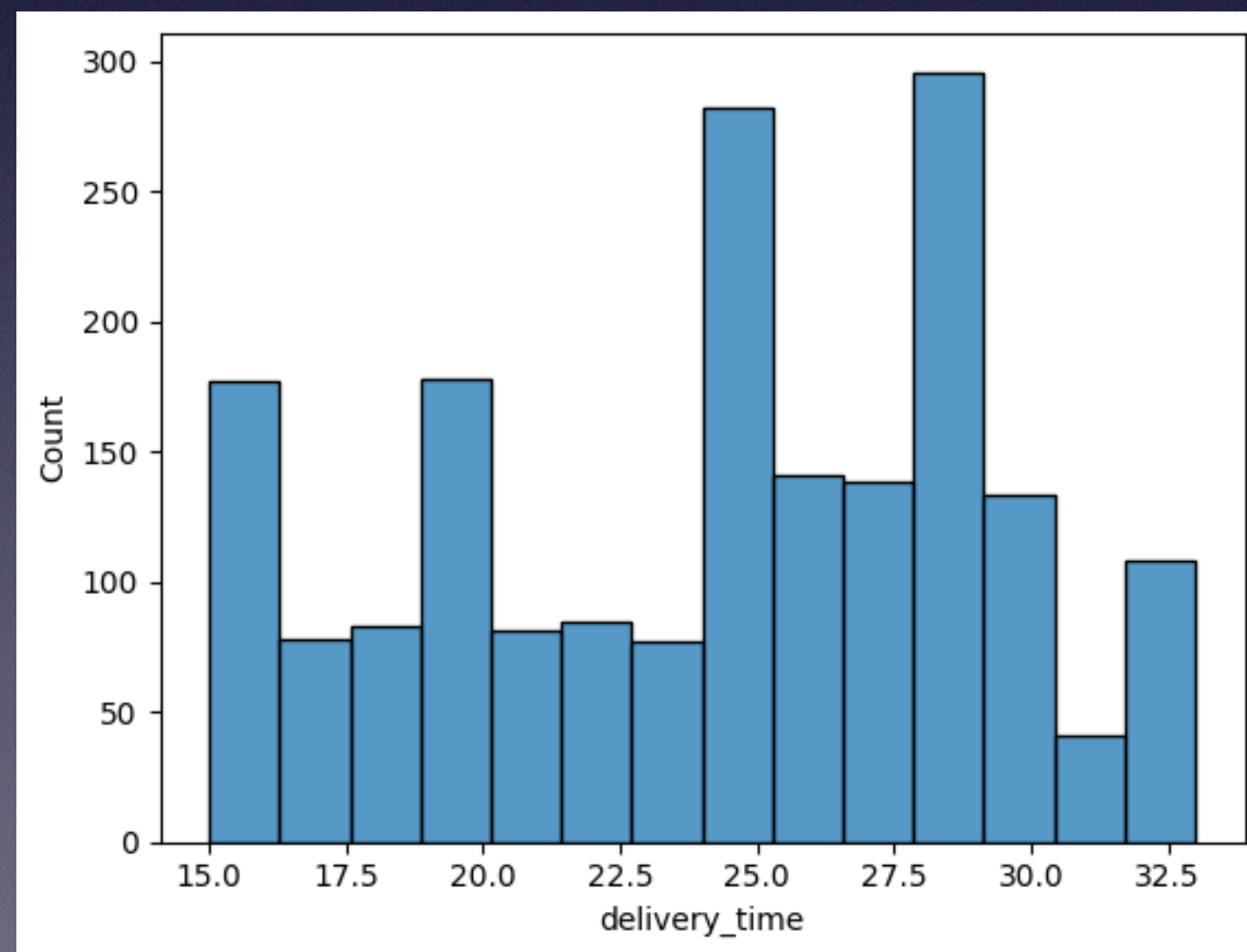
Food Preparation Time

In the 2 charts below we can see how long orders take to be prepared. We see that the minimum time record to prepare food is ~20 minutes and the maximum is ~35 minutes. We also see 25% of orders are prepared in ~23 minutes or less, 50% of the orders take ~27 minutes to prepare and 75% of the orders take ~31 minutes to prepare.



Delivery Time

Below we can see the delivery time of the orders. The minimum time it takes for food to deliver is ~15 minutes and the maximum time is just over 32.5 minutes. We do see 50% of orders are delivered in 25 minutes or less and 75% of orders are delivered in ~28 minutes or less.



Univariate Analysis Cont. #7 & #8

We see the top 5 restaurants are:

- Shake shack with 219 orders
- The Meatball Shop with 132 orders
- Blue Ribbon Sushi with 119 orders
- Blue Ribbon Fried Chicken with 96 orders
- Parm with 68 orders

Most popular cuisine on the weekends:

- American with 169 orders
- Japanese with 135 orders
- Italian with 91 orders
- Chinese with 52 orders
- Mexican with 24 orders

Univariate Analysis Cont. #9,#10 & #11

- Looking further into the data we found that the number of total orders that cost above \$20 is 555 and that is 29.24% of all orders
- The mean order delivery time is 24.16 minutes
- The top 5 most frequent customers who received the 20% discount vouchers have the customer ID's:
 - 52832 (13 orders)
 - 47440 (10 orders)
 - 83287 (9 orders)
 - 250494 (8 orders)
 - 259341 (7 orders)

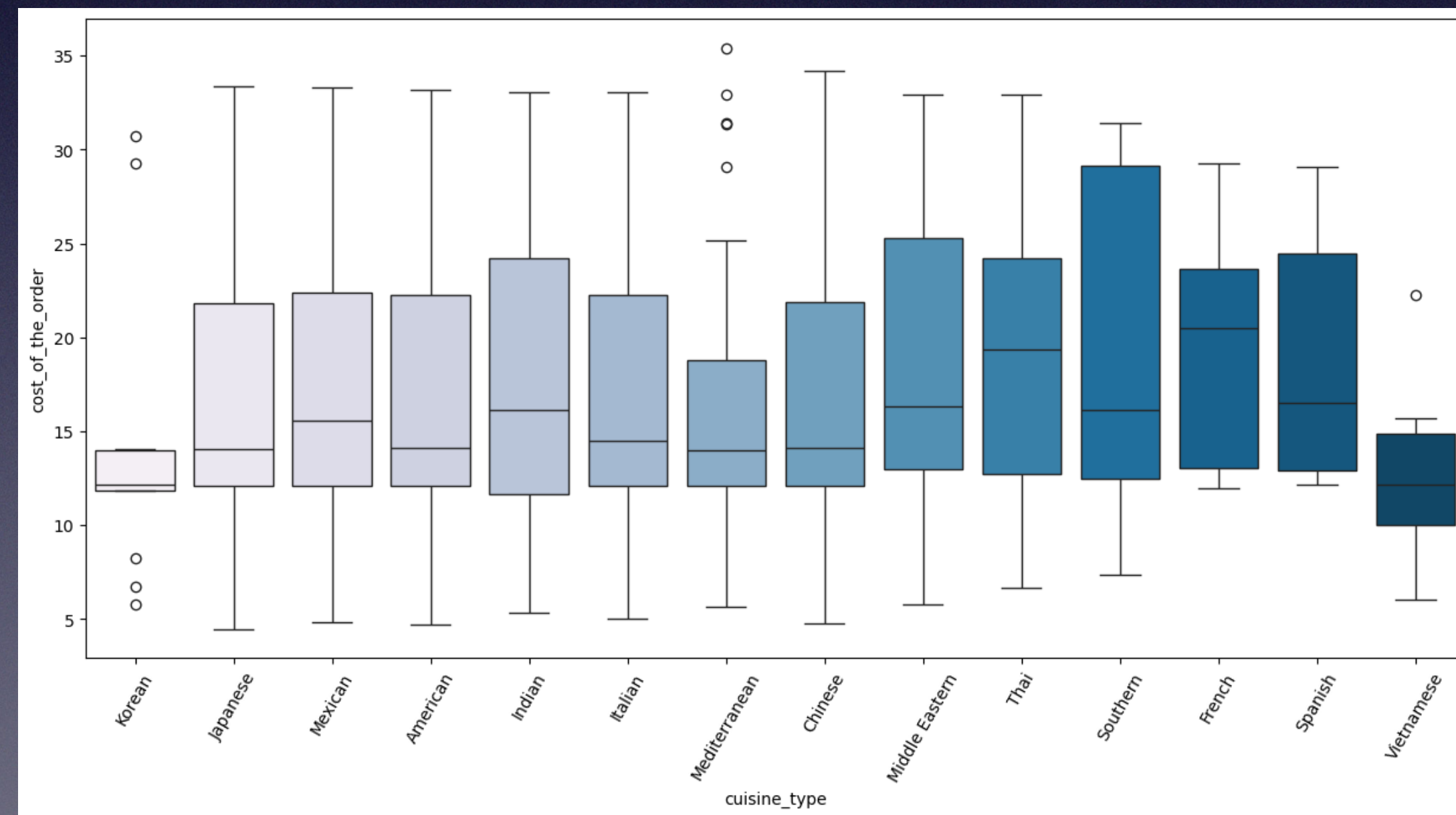
Multivariate Analysis

We will perform a multivariate analysis to explore the relationship between important variables in the dataset. These variables are:

- Cuisine vs Cost_of_the_order
- Cuisine vs Food_Preparation_time
- Day_of_the_Week vs Delivery_time
- Rating vs Delivery_time
- Rating vs Food_Preparation_time
- Rating vs Cost_of_the_order

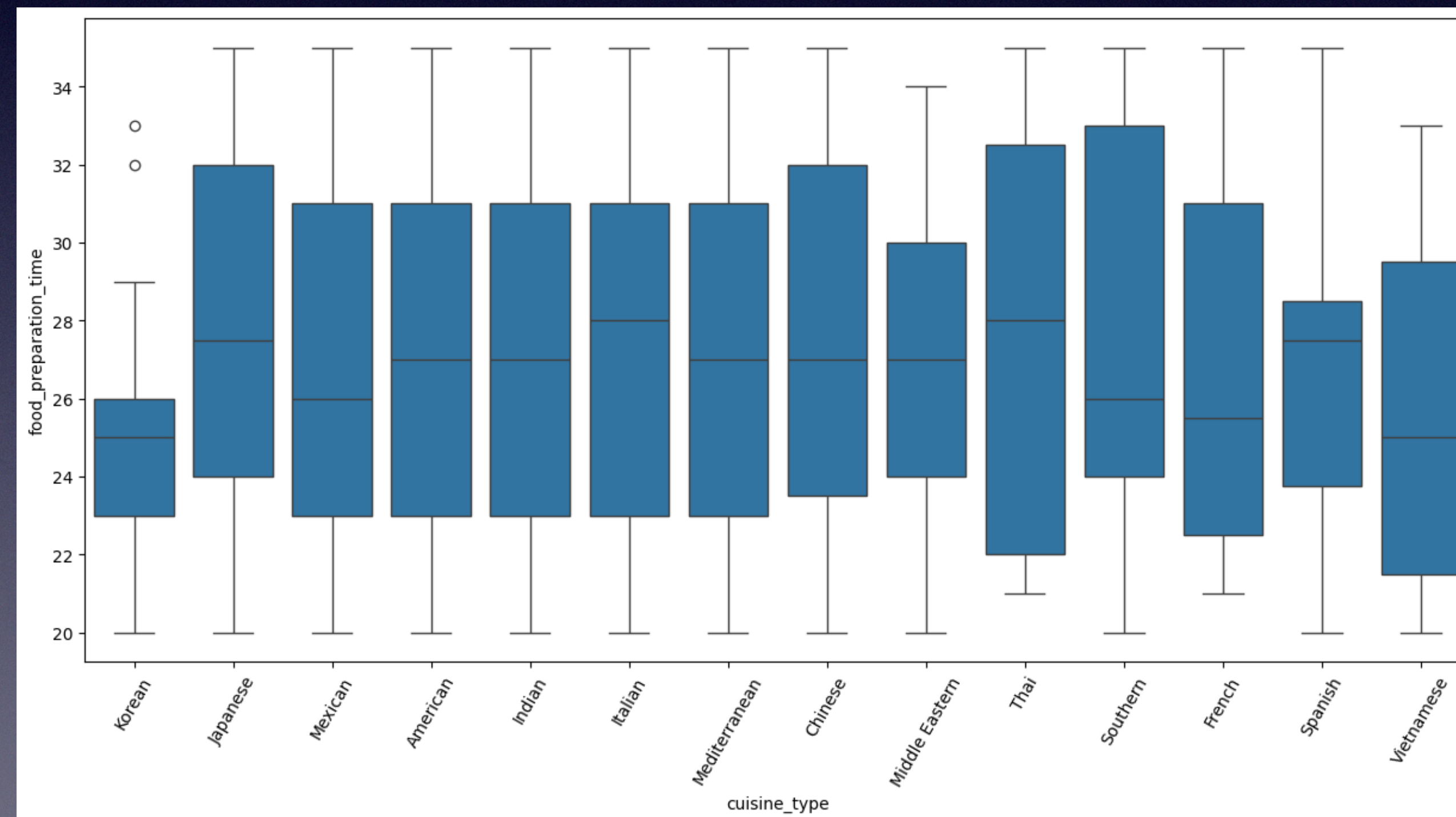
Multivariate Analysis of Cuisine vs Cost of the Order

Some observations we can make in the graph are Korean and Vietnamese food have the lowest median cost out of all cuisines. French and Thai food have the highest median costs out of all cuisines. Most of the cuisines costs range from ~\$13 - ~\$20.



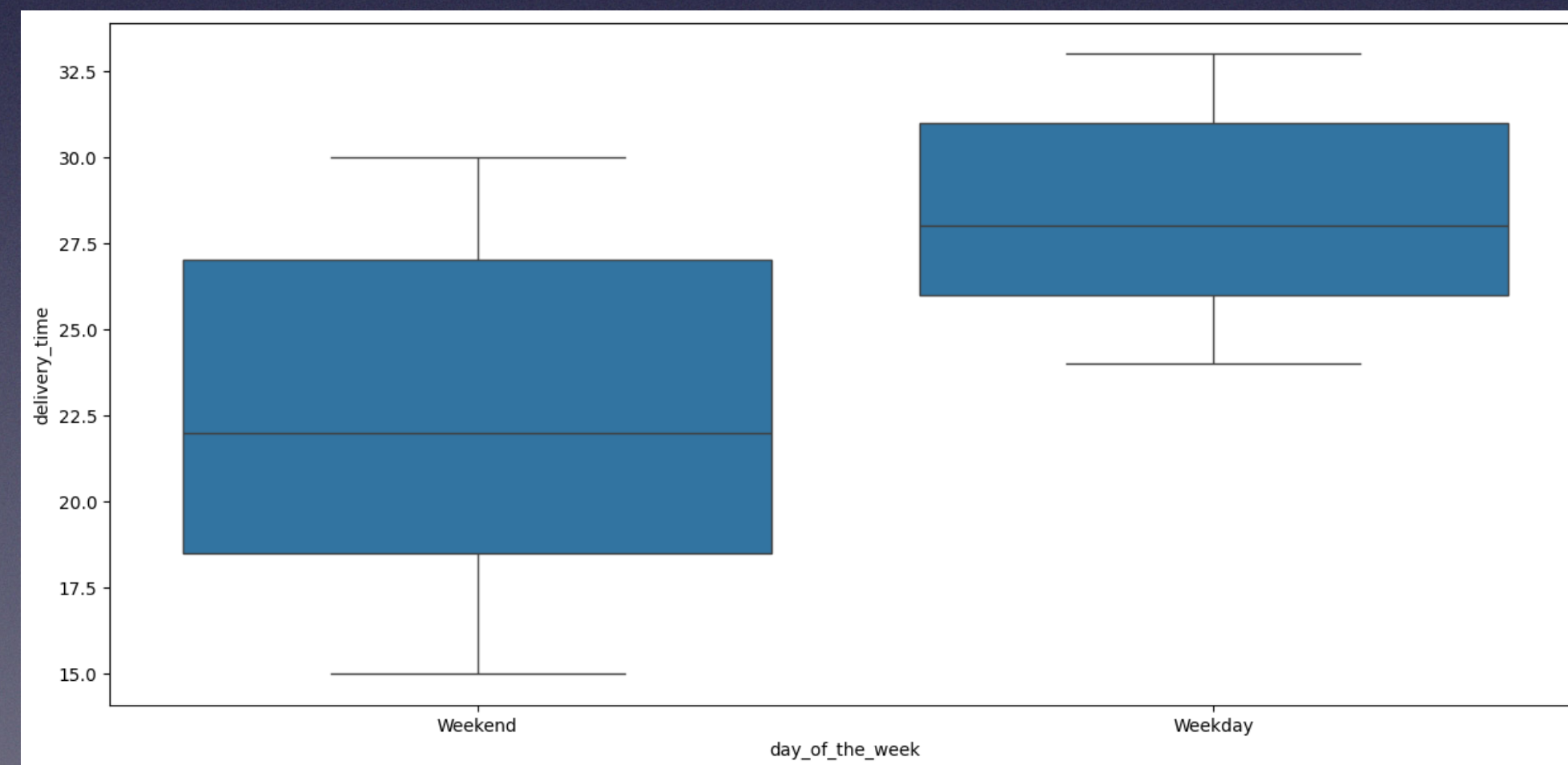
Cuisine vs Food Preparation Time

Some of the observations we can make from this graph is that Korean and Vietnamese have the lowest median preparation time. Thai and Italian food have the highest median preparation time. The majority of the median preparation times in the data are in between ~25 - ~28 minutes.



Day of the Week vs Delivery time

Looking at the graph below we see delivery times are quicker on the weekends than the weekday. The minimum time an order was delivered on the weekend was 15 minutes compared to ~25 minutes on the weekday. Overall customers receive their orders faster on weekends.



Revenue Generated by the Restaurants

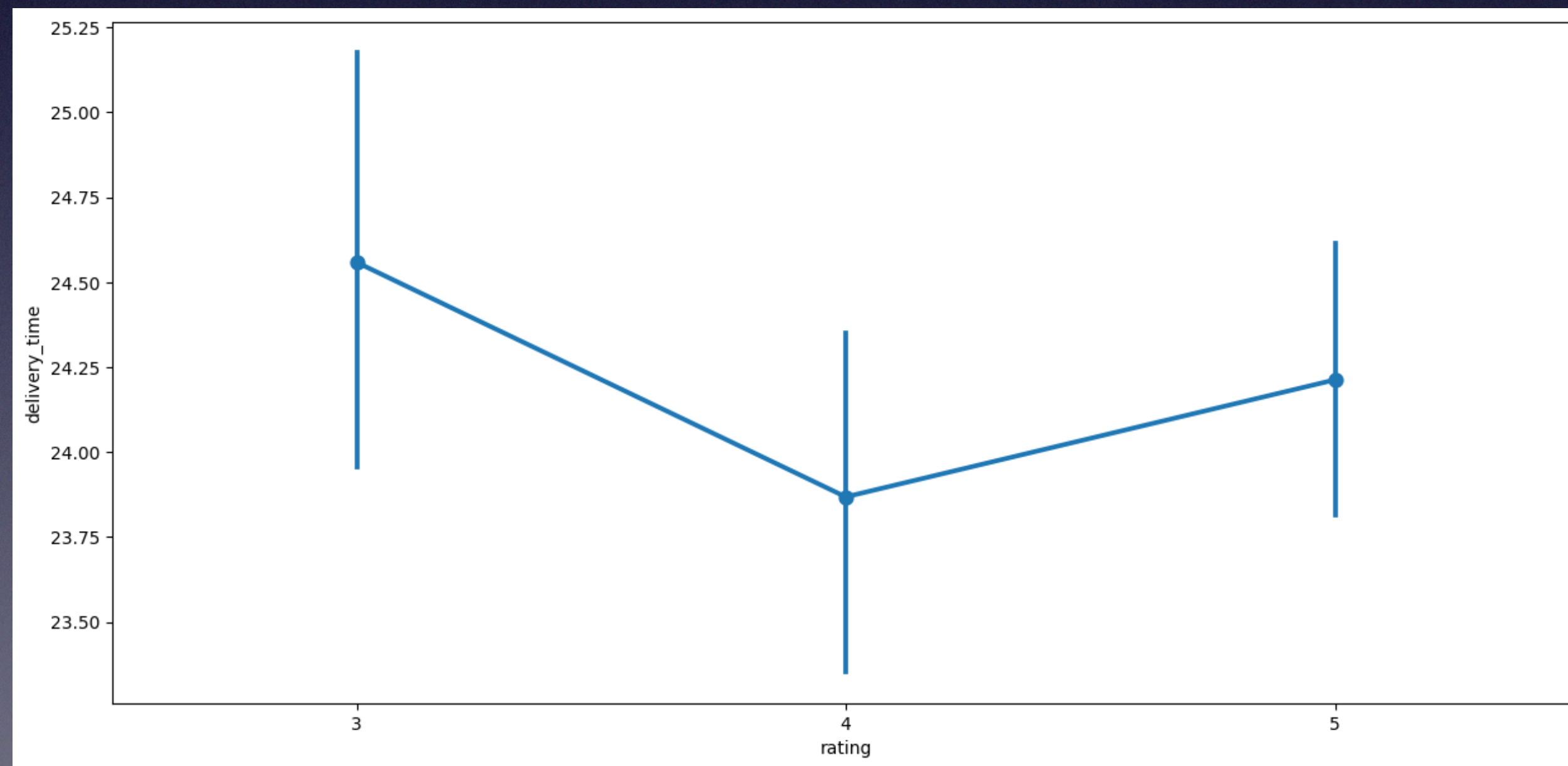
We will look at the revenue generated from the top 5 restaurants:

- Shake Shack = \$3,579.53
- The Meatball Shop = \$2,145.21
- Blue Ribbon Sushi = \$1,903.95
- Blue Ribbon Fried Chicken = \$1,662.29
- Parm = \$1,112.76

We got this data by summing up the cost of the orders placed at these restaurants. Shake Shack has generated the most revenue and all top 5 restaurants are generating over \$1,000.

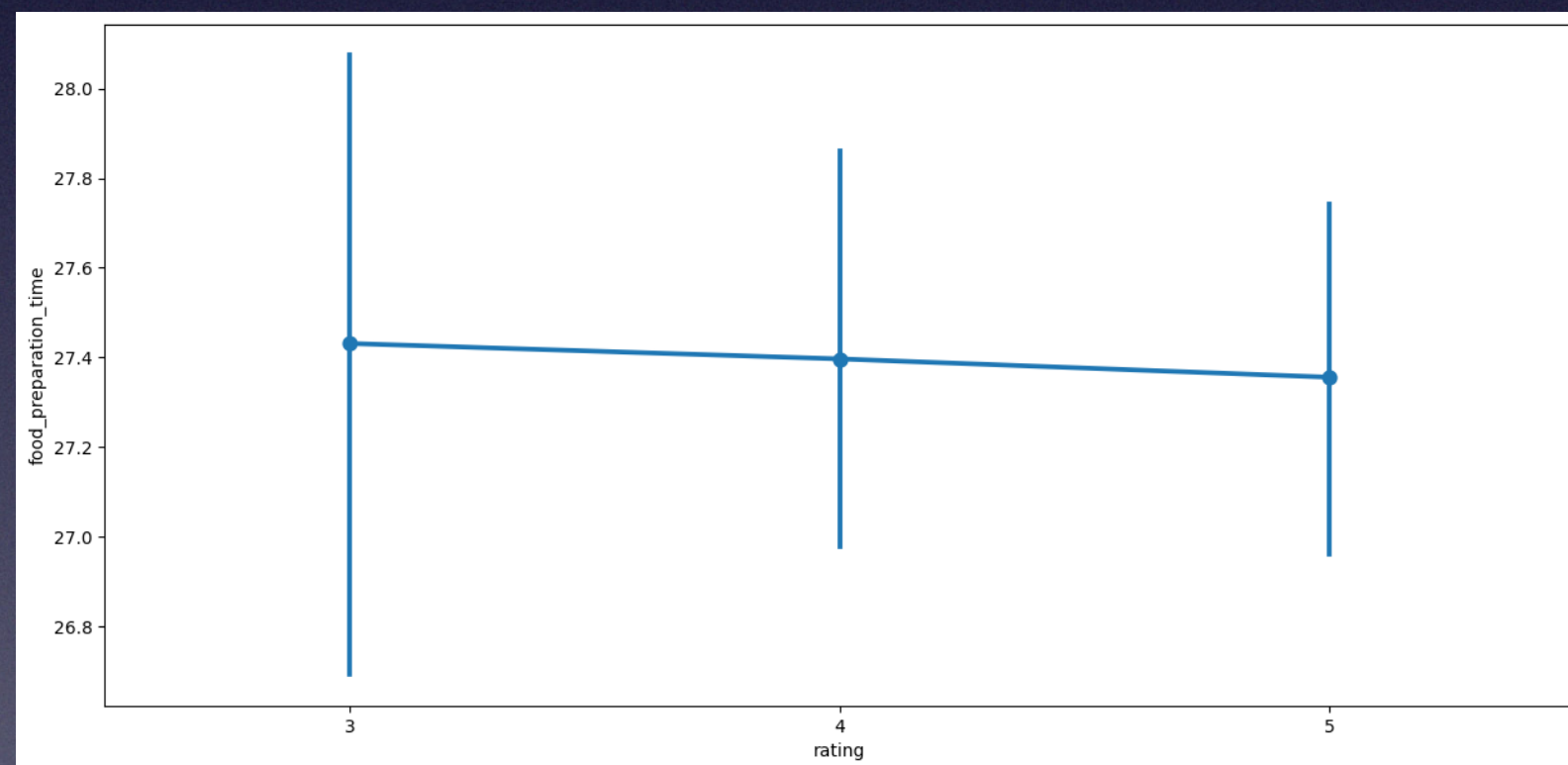
Rating vs Delivery Time

We see that as the rating increased from 3-4 the mean delivery time was lower. It slightly increased when the rating went from a 4-5. The Average delivery times were in-between 24 - 24.75 minutes.



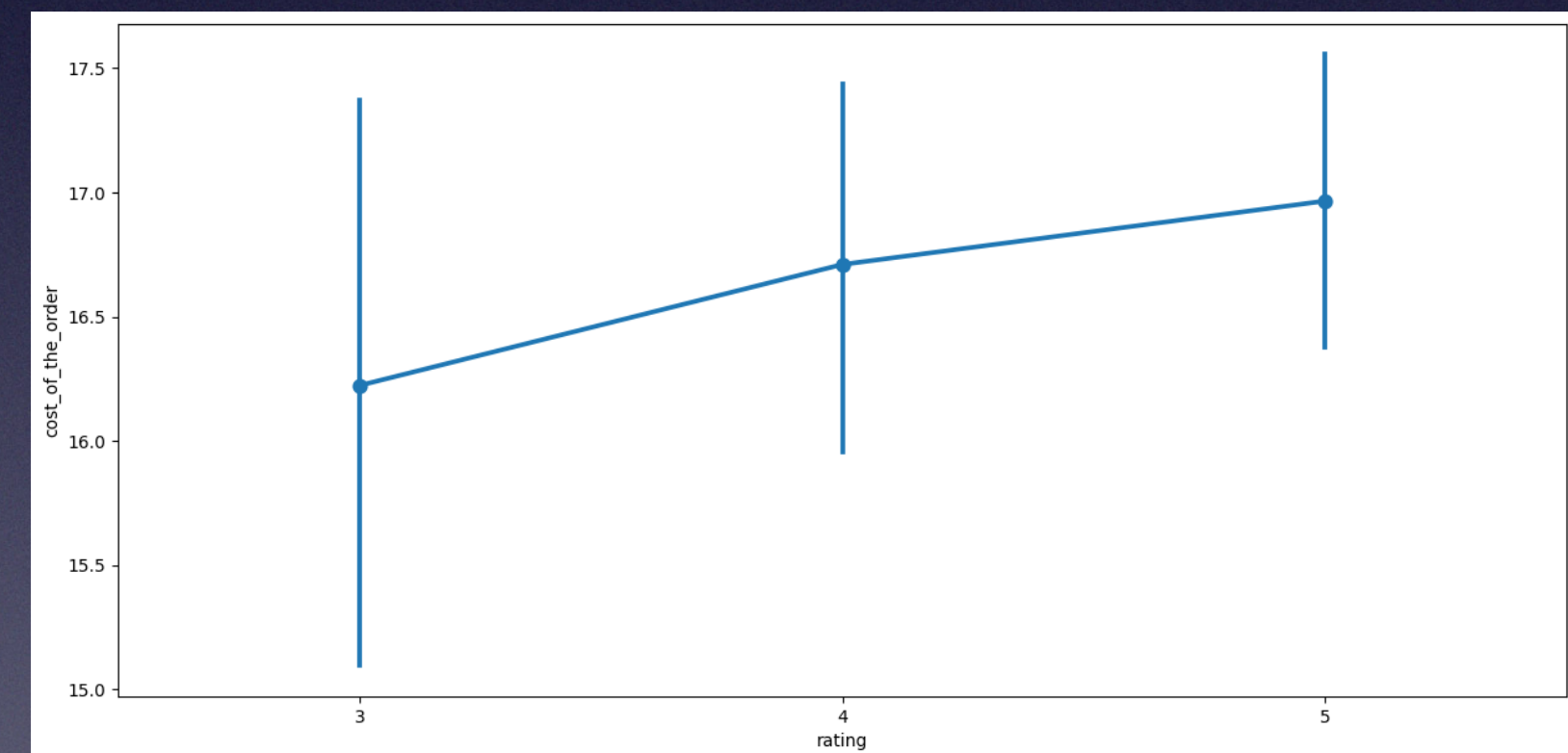
Rating vs Food Preparation Time

We can see as the food_preparation_time goes down the rating goes up. It is a negative correlation.



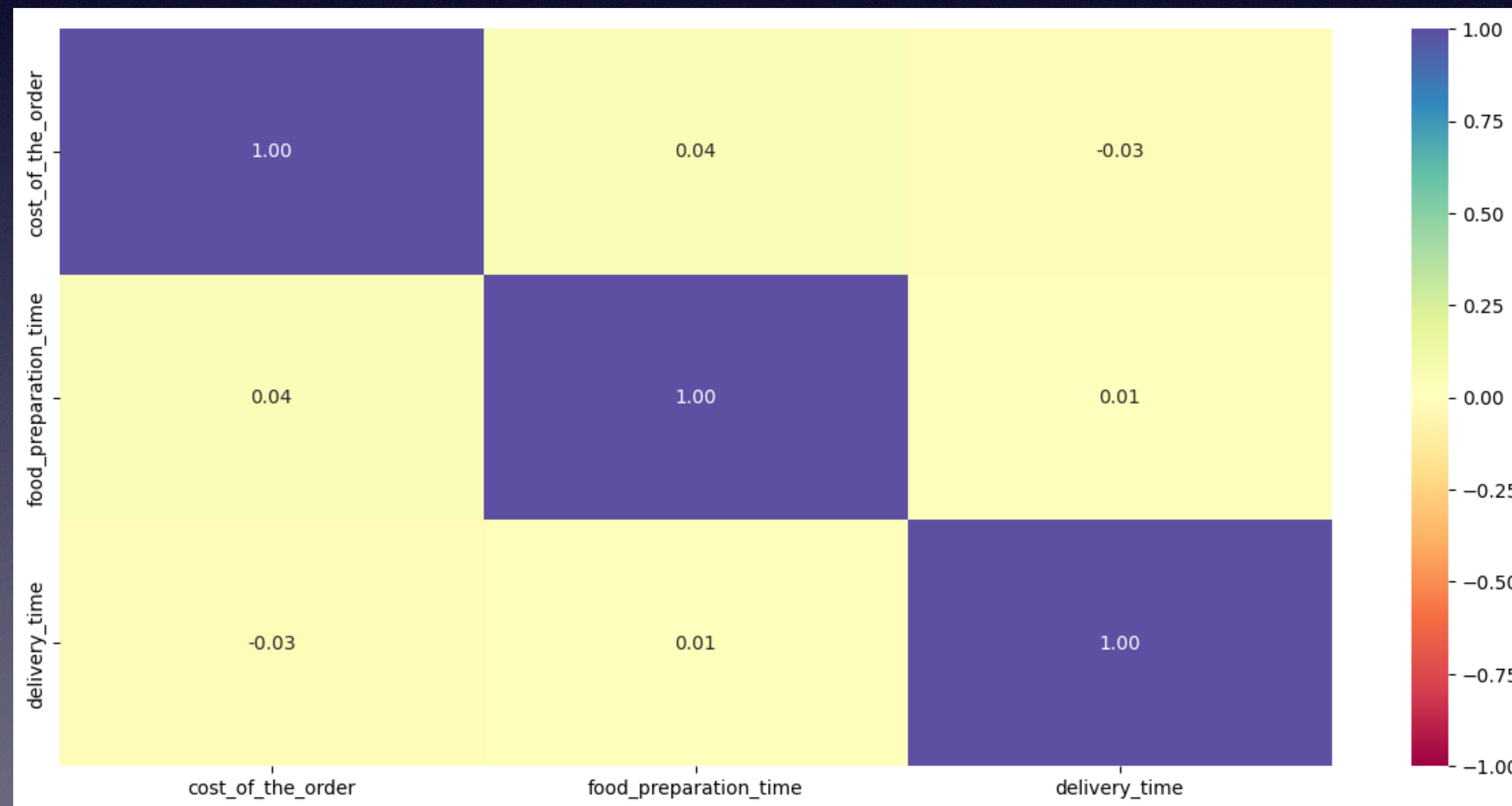
Rating vs Cost of the Order

We can see as the cost_of_the_order goes up the rating also increases. These variables have a positive correlation.



Correlation among Variables

We see on the Heat map that all correlations are very small. Food_preparation_time & cost_of_the_order have the highest correlation of all variables listed.



Multivariable Analysis Questions 13 & 14

The company wants to provide a promotional offer in the advertisement of the restaurants. The condition they have set is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4.

The restaurants that qualify are:

- Shake Shack - 133 ratings with an average rating of 4.278195
- The Meatball Shop - 84 ratings with an average rating of 4.511905
- Blue Ribbon Sushi - 73 ratings with an average rating of 4.219178
- Blue Ribbon Fried Chicken - 64 ratings with an average rating of 4.328125

These are the 4 restaurants that will receive the promotional offer.

The company also charges restaurants 25% on orders having cost greater than 20 dollars and 15% on orders having cost greater than 5 dollars. The net revenue generated by the company across all orders is around \$6,166.3.

Multivariable Analysis Questions 15 & 16

The company wants to analyze the total time required to deliver the food. The percentage of orders taking more than 60 minutes is 10.54%.

The company also wants to analyze the delivery time of the orders on weekdays and weekends. The mean delivery time is:

- Around 28 minutes on weekdays
- Around 22 minutes on weekends

Thank You!